

Sakshi Vedi

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence & Machine Learning candidate at **IIIT Lucknow** with strong expertise in **Machine Learning, Deep Learning, Computer Vision, and Generative AI**. Hands-on experience in building **LLM-based agent systems, RAG pipelines**, and **real-time computer vision applications**. Proficient in **Python-driven ML workflows**, model deployment using **Streamlit**, and applying **MLOps principles** for reproducible and scalable AI solutions.

EDUCATION

MSc (IT) – Artificial Intelligence & Machine Learning — IIIT Lucknow (CGPA: 8.62/10) — Present

BSc – Mathematics — Maharshi Dayanand Saraswati University, Ajmer (74.27%) — Completed

EXPERIENCE

- **Data Analyst Intern** Apr 2026 – Jun 2026 Obel System Pvt. Ltd.
 - Worked on the **Rahat** project under the Government of Uttar Pradesh, managing large-scale government portal data, performing data validation, cleaning, and ensuring consistency across records.
 - Generated reports, monitored data quality, resolved data discrepancies, and collaborated with cross-functional teams to support daily operations and decision-making.
- **AI Intern** Apr 2026 – May 2026 TaskNova
 - Developed backend services using **FastAPI** and integrated **LLM-based conversational AI**, implementing emotion detection, crisis detection, and personalized context management modules.
 - Built REST APIs, worked with **Supabase (PostgreSQL)**, **Celery**, **Redis**, Docker, and voice AI pipelines (Speech-to-Text/Text-to-Speech) for scalable AI applications.
- **AI/ML Engineer Intern** Feb 2026 – Mar 2026 Gaara AI
 - Developed a real-time AI pose detection and correction system using **MediaPipe Pose**, extracting 33 body landmarks and calculating joint angles for posture analysis.
 - Designed pose validation and correction modules, optimized inference for low-latency performance, and evaluated multiple pose estimation frameworks including MediaPipe, MoveNet, and YOLO Pose.
- **Climate Data Science Volunteer** Jul 2025 – Jan 2026 IIIT Lucknow
 - Built an **XGBoost**-based flood prediction model using hydrological and meteorological data, applying preprocessing, feature engineering, and **SMOTE** to handle class imbalance.
 - Achieved **77% accuracy** in predicting flood risk across multiple river basins, supporting early warning systems and disaster preparedness initiatives.

TECHNICAL PROJECTS

- **Multi Agent AI Trip Planner** [\[Github\]](#)
 - Built a **multi-agent AI travel planner** using **LangGraph**, LLMs, and real-time APIs to generate personalized itineraries with parallel agent execution.
 - Developed an interactive **Streamlit** interface with PDF/email export, reducing itinerary generation time from **90s to 80s**.
- **RAG-Powered Chatbot** [\[Github\]](#)
 - Built a **RAG-based chatbot** using local embeddings, semantic search, and **Groq LLaMA** to answer queries from PDFs and Markdown documents.
 - Implemented vector similarity retrieval, source citations, and a **Dockerized Streamlit** deployment to deliver fast, grounded responses.
- **Eye Drowsiness Detection System** [\[Github\]](#)
 - Developed a real-time driver drowsiness detection system using **OpenCV**, facial landmarks, and Eye Aspect Ratio (EAR) analysis.
 - Implemented an audio alert mechanism for prolonged eye closure and optimized the pipeline for low-latency CPU execution.
- **Fine-Tuned LLaMA 2** [\[Github\]](#)
 - Fine-tuned **LLaMA 2** using **QLoRA**, Hugging Face Transformers, and PyTorch on domain-specific datasets for improved response quality.
 - Built end-to-end preprocessing, training, and inference pipelines with optimized memory-efficient model deployment.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, LaTeX, PowerShell, Shell Scripting
- **Machine Learning, Deep Learning & Data Science:** Supervised Learning, Unsupervised Learning, Random Forest, XGBoost, LightGBM, CNNs, RNNs, LSTMs, Transformers, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, OpenCV, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), NLP, OCR, Image-based Prediction Systems, Model Evaluation
- **Generative AI & AI Engineering:** Large Language Models (LLMs), Generative AI, RAG Pipelines, Agentic AI, LangChain, LangGraph, Prompt Engineering, Fine-tuning (PEFT, LoRA, QLoRA), Model Quantization, Hugging Face, OpenAI APIs, MCP Server, Edge TTS, AI Response Evaluation & Validation, LLM-as-a-Judge Frameworks, Deterministic & Probabilistic AI Systems, Harness Engineering, Modular AI Workflow Design
- **Data Engineering, MLOps & Backend:** Docker, DVC, CI/CD Pipelines, MLOps, Model Deployment, FastAPI, MongoDB, MySQL, ChromaDB, Vector Databases, REST API
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, Streamlit, .NET
- **Soft Skills:** Analytical Thinking, Research Orientation, Technical Communication, Problem Solving